Contents

1 Abstract **1**

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We evaluate the vibrational theory of olfaction, which posits that insects can detect infrared electromagnetic radiation emitted by semiochemicals alongside or in addition to molecular binding mechanisms. Using morphological measurements, spectroscopic analyses, neural latency comparisons, and tested computational models implemented in `src/`, we assess the plausibility and testable predictions of this hypothesis. Our thesis is that insects achieve rapid, long-range, and frequency-specific chemosensation by leveraging infrared transmission windows and resonant biological structures, and that this mechanism yields falsifiable, quantitative predictions reproduced by our open tests and figures.

Key results: (i) Sensilla geometry aligns with predicted resonant wavelengths derived from dielectric waveguide models; (ii) published olfactory receptor neuron latencies ($\approx\ 1 - -5ms)areconsistentwithrapid, non - diffusion - limiteddetection; (iii)cuticularhydrocarbon(CHC)spectralpeaksfallwithinmodeledatmospherictransmis$ $-5m, 8--14m, 17--25m)using$`src/core.calculate_atmospheric_transmission`$(validatedin$`tests/te`

Framework: We integrate resonant cavity and waveguide theory with CHC spectroscopy (`src/spectroscopy.py`) and sensilla morphology (`src/sensilla.py`). Implementations are covered by targeted unit tests (e.g., `tests/test_core.py::TestAtmosphericTransmission`, `tests/test_sensilla.py::TestSensillaAnalysis`, `tests/test_spectroscopy_analysis.py::TestAnalyzeChcSpectra`). Each claim is paired with a runnable example (fixed seed 42) and a generated figure path reported to `output/figures/`.

Scope and implications: Findings provide falsifiable predictions for wavelength tuning, behavioral IR-only responses, and neural responses to IR stimulation. If validated, these mechanisms have implications for sensory biology and biomimetic sensing, and they delineate minimal falsification tests enumerated in Discussion.

Reproducibility: Analyses are deterministic (fixed seeds) and reproducible via the unified pipeline: run tests for 100% coverage, regenerate figures with `scripts/generate_research_figures.py`, and compile the manuscript through `repo_utilities/render_pdf.sh`. Method cross-links are listed in **??** and the Symbols/Glossary.